Preventing the First Weapon of Mass Extinction:

Addressing Artificial Intelligence for Bioweapons Development

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Abstract

The COVID-19 pandemic demonstrated the devastating destabilization of pathogens, exemplifying attention for the threat of bioweapons. With artificial intelligence's increasing dual-use capabilities, the barriers to develop bioweapons have been removed, potentially transforming this weapon of mass destruction into a weapon of mass extinction. Most research focuses on anticipating future threats of AI, such as creating pathogens that target specific ethnic groups or redesigning known pathogens to circumvent existing interventions. Research on current capabilities finds that AI can design and develop novel pathogens with little to no expertise. To combat bioweapons, scholars have applied nuclear deterrence theory and formed three camps: deterrence by punishment, denial, and dissuasion. While dissuasion was the more realistic strategy in the past, the reality of AI today has left deterrence by denial, which necessitates robust biodefense capabilities, to be the only effective option. There is little research on AI for biodefense with the sole focus on institutional or system level applications that still allow for significant response latency. Shifting towards AI that operates on a direct patient level as frontline biodefense is essential to enter a new phase of biosecurity.

Introduction

After the advent of nuclear weapons and their devastating effects, global attention remains on weapons of mass destruction (WMD).[1] Among these is biological weapons or bioweapons, which is defined by the World Health Organization as "microorganisms like virus, bacteria or

[1] R. Everett Langford, *Introduction to Weapons of Mass Destruction: Radiological, Chemical, and Biological* (John Wiley & Sons, 2004).

fungi, or toxic substances produced by living organisms that are produced and released deliberately to cause disease and death in humans, animals or plants."[2] While there exists a long history of bioweapons, attention exemplified after the societal destabilization that ensued with COVID-19, so-called "nature's virtually perfect biological weapon."[3] Because bioweapons are manmade pathogens capable of rapidly spreading, continually mutating, burdensome symptoms, they are an unmatched threat. In humanity's history, smallpox is the only human disease to have been eradicated.[4] Thus, history tells us once a bioweapon unleashes itself to the world, it will most definitely stay forever.

Modern international regulations regarding bioweapons began with the 1925 Geneva Protocol, which only prohibited the use of bioweapons.[5] To extend upon this, countries signed the Biological Weapons Convention (BWC) treaty in 1972 to prohibit the development, production, and stockpiling of biological and toxic weapons, being the first multilateral treaty to effectively ban a whole WMD category.[6] However, its effectiveness is extremely limited due to the lack of a cohesive verification protocol mechanism.[7] Unlike with nuclear weapons continually monitored

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[2] World Health Organization, "Biological Weapons," World Health Organization, World Health Organization, accessed October 31, 2025, https://www.who.int/health-topics/biological-weapons.
[3] Giuseppe Lippi et al., "COVID-19: Unravelling the Clinical Progression of Nature's Virtually Perfect Biological Weapon," *Annals of Translational Medicine* 8, no. 11 (2020): 693, https://doi.org/10.21037/atm-20-3989.
[4] Hervé Bazin, "Introduction," in *The Eradication of Smallpox: Edward Jenner and the First and Only Eradication of a Human Infectious Disease*, ed. Hervé Bazin (Academic Press, 2000), https://doi.org/10.1016/B978-012083475-4/50003-5; Donald A. Henderson, "A History of Eradication—Successes, Failures, and Controversies," *The Lancet* 379, no. 9819 (2012): 884–85, https://doi.org/10.1016/S0140-6736(12)60381-X.
[5] "Protocol for the Prohibition of the Use in War of Asphyxiating, Poisonous or Other Gases, and of Bacteriological Methods of Warfare (Geneva Protocol)," U.S. Department of State, U.S. Department of State, June 17, 1925, //2009-2017.state.gov/t/isn/4784.htm.
[6] Liang Huigang et al., "Development of and Prospects for the Biological Weapons Convention," *Journal of Biosafety and Biosecurity* 4, no. 1 (2022): 50–53, https://doi.org/10.1016/j.jobb.2021.11.003; "Convention on the Prohibition of the Development, Production and Stockpiling of Bacteriological (Biological) and Toxin Weapons and on Their Destruction," United Nations, United Nations Office for Disarmament Affairs, April 10, 1972, https://treaties.unoda.org/t/bwc.
[7] Huigang et al., "Development of and Prospects for the Biological Weapons Convention"; Marie Isabelle Chevrier, "From Verification to Strengthening Compliance: Prospects and Challenges of the Biological Weapons Convention," *Politics and the Life Sciences* 14, no. 2 (2016): 209–19, https://doi.org/10.1017/S0730938400019109; Kristen Paris,

by the International Atomic Energy Agency, no such organization exists for bioweapons, leaving the world wholly unprepared to prevent or respond to an attack.

Immense strides in biotechnology over the last decade have greatly benefited humanity while amplifying the bioweapons threat. Next generation sequencing rapidly allows entire genomes to be sequenced in a day.[8] CRISPR-Cas9 enables precise genome editing.[9] mRNA vaccines have high efficacy with rapid development timelines.[10] If used responsibly, advances can save and improve billions of lives, from precision medicine to more nutritious crops.[11] However, malicious actors can leverage the same benefits to destroy lives.

With the increasing dual-use capabilities of artificial intelligence (AI), the world is as close as ever to actual bioweapons being developed and released. In 2022, former Google Chief Executive Officer Eric Schmidt warned the biggest issue in AI is biological warfare.[12] Biochemist Erwin Chargaff even warned in 1978 when genetic modification first emerged that genetic engineering poses a greater danger to the world than nuclear technology.[13] As AI continues to disrupt, that same warning carries new meaning. AI has the potential to transform bioweapons from a WMD to a WME: a weapon of mass extinction.

"The Expansion of the Biological Weapons Convention: The History and Problems of a Verification Regime," *Houston Journal of International Law* 24, no. 3 (2002), https://heinonline.org/HOL/P?h=hein.journals/hujil24&i=518.

[8] Sam Behjati and Patrick S Tarpey, "What Is next Generation Sequencing?," *Archives of Disease in Childhood. Education and Practice Edition* 98, no. 6 (2013): 236–38, https://doi.org/10.1136/archdischild-2013-304340.

[9] Rodolphe Barrangou and Jennifer A. Doudna, "Applications of CRISPR Technologies in Research and Beyond," *Nature Biotechnology* 34, no. 9 (2016): 933–41, https://doi.org/10.1038/nbt.3659.

[10] Norbert Pardi et al., "mRNA Vaccines — a New Era in Vaccinology," *Nature Reviews Drug Discovery* 17, no. 4 (2018): 261–79, https://doi.org/10.1038/nrd.2017.243.

[11] Saurav Das et al., "Role of Biotechnology in Creating Sustainable Agriculture," *PLOS Sustainability and Transformation* 2, no. 7 (2023): e0000069, https://doi.org/10.1371/journal.pstr.0000069.

[12] Ryan Lovelace, "AI-Powered Biological Warfare Is 'biggest Issue,' Former Google Exec Eric Schmidt Warns - Washington Times," The Washington Times, The Washington Times, September 12, 2022, https://www.washingtontimes.com/news/2022/sep/12/ai-powered-biological-warfare-biggest-issue-former/.

[13] Erwin Chargaff, *Heraclitean Fire: Sketches from a Life Before Nature* (1978).

In this chapter, I will first review the existing literature on AI used for bioweapons creation, including anticipated threats of AI and current capabilities. Then, I will focus on deterrence based strategies against bioweapons and relevance in the present world. Finally, I will evaluate work done on AI for biodefense.

AI for Offensive Bioweapons

ANTICIPATING THREATS OF AI FOR BIOWEAPONS DEVELOPMENT

The vast majority of research at the intersection of AI with biosecurity examines the dual-use risk of AI with significant attention on what AI can potentially do in the future.[14] Such emerging risks include development of genetically modified organisms as bioweapons, genetically programmed nanobots that attack specific anatomical features, pathogens that target susceptible regions of the human genome, microorganisms that attack crops and critical systems, neurotechnological weapons that influence neurological functions, and bioweapons that target specific ethnic groups.[15] Other concerns include undermining existing biosecurity and biodefense measures, such as known toxins and pathogens redesigned to bypass nucleic acid synthesis

[14] Doni Bloomfield et al., "AI and Biosecurity: The Need for Governance," *Science* 385, no. 6711 (2024): 831–33, https://doi.org/10.1126/science.adq1977; Mengdi Wang et al., "A Call for Built-in Biosecurity Safeguards for Generative AI Tools," *Nature Biotechnology* 43, no. 6 (2025): 845–47, https://doi.org/10.1038/s41587-025-02650-8.
[15] Renan Chaves de Lima et al., "Artificial Intelligence Challenges in the Face of Biological Threats: Emerging Catastrophic Risks for Public Health," *Frontiers in Artificial Intelligence* 7 (May 2024), https://doi.org/10.3389/frai.2024.1382356; James Giordano and Rachel Wurzman, "Neurotechnologies as Weapons in National Intelligence and Defense: An Overview," *Synesis: A Journal of Science, Technology, Ethics, and Policy* 2, no. 1 (2011): T55–71; Engineering National Academies of Sciences et al., "AI-Enabled Biological Design and the Risks of Synthetic Biology," in *The Age of AI in the Life Sciences: Benefits and Biosecurity Considerations* (National Academies Press (US), 2025), https://www.ncbi.nlm.nih.gov/books/NBK614591/.

screening, resist medical interventions, and circumvent biosurveillance systems.[16] Essentially, AI makes the worst case possible scenarios more possible.

Particular concern surrounds AI-enabled biological design tools (BDTs), which are technologies trained on biological data for biological applications.[17] One of the latest is Evo 2, the largest biological AI foundation model as of late 2025 that is said to "read, write, and think in the language of nucleotides."[18] Although Evo 2 is trained on a diverse dataset of 128,000 genomes, it is currently incapable, along with other BDTs, of designing complex pathogens.[19] While the state-of-the-art BDTs can design simple molecules, the National Academies of Sciences, Engineering, and Medicine believed in April 2025 that actual self-replicating pathogens are orders of magnitude more difficult due to limited viral sequence data available (more on this later).[20] Regardless, one can imagine the danger of a future advanced BDT, granting the opportunity to design a bioweapon at the click of a button.

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[16] National Academies of Sciences et al., "AI-Enabled Biological Design and the Risks of Synthetic Biology"; Sean Ekins et al., "Generative Artificial Intelligence-Assisted Protein Design Must Consider Repurposing Potential," *GEN Biotechnology* 2, no. 4 (2023): 296–300, https://doi.org/10.1089/genbio.2023.0025; David Baker and George Church, "Protein Design Meets Biosecurity," *Science* 383, no. 6681 (2024): 349–349, https://doi.org/10.1126/science.ado1671; Bruce J. Wittmann et al., "Toward AI-Resilient Screening of Nucleic Acid Synthesis Orders: Process, Results, and Recommendations," preprint, bioRxiv, December 4, 2024, 2024.12.02.626439, https://doi.org/10.1101/2024.12.02.626439.

[17] National Academies of Sciences et al., "AI-Enabled Biological Design and the Risks of Synthetic Biology"; Sarah Morgan, "The Double-Edged Sword: Opportunities and Risks of AI in Biosecurity," *Georgetown Security Studies Review*, November 15, 2024, https://gssr.georgetown.edu/the-forum/topics/biosec/the-double-edged-sword-opportunities-and-risks-of-ai-in-biosecurity/; Georgia Adamson and Gregory C. Allen, *Opportunities to Strengthen U.S. Biosecurity from AI-Enabled Bioterrorism: What Policymakers Should Know*, August 6, 2025, https://www.csis.org/analysis/opportunities-strengthen-us-biosecurity-ai-enabled-bioterrorism-what-policymakers-should.

[18] "AI Can Now Model and Design the Genetic Code for All Domains of Life with Evo 2 | Arc Institute," Arc Institute, Arc Institute, February 19, 2025, https://arcinstitute.org/news/evo2; Adamson and Allen, *Opportunities to Strengthen U.S. Biosecurity from AI-Enabled Bioterrorism*.

[19] National Academies of Sciences et al., "AI-Enabled Biological Design and the Risks of Synthetic Biology"; Adamson and Allen, *Opportunities to Strengthen U.S. Biosecurity from AI-Enabled Bioterrorism*.

[20] National Academies of Sciences et al., "AI-Enabled Biological Design and the Risks of Synthetic Biology"; Adamson and Allen, *Opportunities to Strengthen U.S. Biosecurity from AI-Enabled Bioterrorism*.

CURRENT CAPABILITIES OF AI FOR BIOWEAPONS DEVELOPMENT

Prior research has shown that many of the conceived threats are becoming reality. For example, one study examined the biosecurity challenges presented by open source AI protein design software.[21] The result 72 AI-engineered toxins that evaded detection by current biosecurity screening tools.[22] Although the team developed patches to improve detection rates, the most frightening part is the publication date of October 2025.[23] Eight days after this publication, *NBC News* revealed its own results of using simple prompts that bypass OpenAI's security rules called "jailbreaks" to trick four of the company's most advanced models to disguise a bioweapon and give steps to make a pathogen target the immune system.[24] This implies that we are only recently understanding the true capabilities of AI for bioweapons as reflected by the exponential growth of publications specifically related to AI and biosecurity in Figure 1.[25] These investigations are the sole ones to study within a biosecurity context what AI can currently do beyond hypotheticals. Due to this limited work, to better understand AI's dual-use risk, it is worthwhile to look at how AI is improving for therapeutic purposes.

As is the nature with AI, previous discoveries even a year ago are moot due to the rapid improvement of quality that enhances capabilities. The greatest advances are in combining existing models to create AI systems. In 2023, Microsoft evaluated the impacts of GPT-4 on scientific discovery, finding that it could provide instructions on how to design an antibody that

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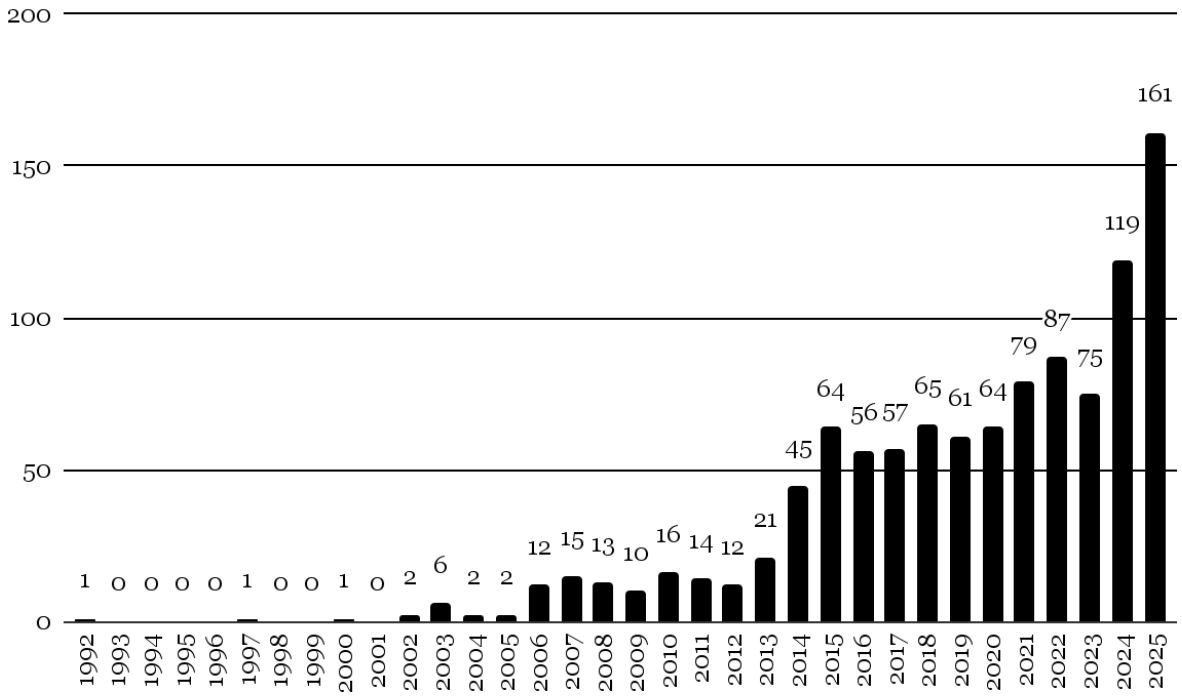
[21] Bruce J. Wittmann et al., "Strengthening Nucleic Acid Biosecurity Screening against Generative Protein Design Tools," *Science* 390, no. 6768 (2025): 82–87, https://doi.org/10.1126/science.adu8578.
[22] Wittmann et al., "Strengthening Nucleic Acid Biosecurity Screening against Generative Protein Design Tools."
[23] Wittmann et al., "Strengthening Nucleic Acid Biosecurity Screening against Generative Protein Design Tools."
[24] Kevin Collier and Jasmine Cui, "ChatGPT Safety Systems Can Be Bypassed to Get Weapons Instructions," NBC News, October 10, 2025, https://www.nbcnews.com/tech/security/chatgpt-safety-systems-can-bypassed-weapons-instructions-rcna225788.
[25] I obtained this data on November 2, 2025 using the following search query on PubMed: (AI OR artificial intelligence) AND biosecurity.

Figure 1. Publication Counts Related to AI and Biosecurity by Year on PubMed, 1992-2025



can bind to the SARS-CoV-2 spike protein and convert an experiment into code to automate a robot specializing in liquids.[26] Later that year, scientists leveraged GPT-4 to develop an AI system for chemical research called Coscientist that designed, planned, and performed experiments utilizing a robotic liquid handler.[27] In July 2025, a team used GPT-4o to create a team of AI agents called the Virtual Lab to enable AI-human research collaboration for interdisciplinary research.[28] This system designed new nanobody binders to SARS-CoV-2 and created a computational nanobody design pipeline.[29]

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[26] Microsoft Research AI4Science and Microsoft Azure Quantum, "The Impact of Large Language Models on Scientific Discovery: A Preliminary Study Using GPT-4," arXiv:2311.07361, preprint, arXiv, December 8, 2023, https://doi.org/10.48550/arXiv.2311.07361.
[27] Daniil A. Boiko et al., "Autonomous Chemical Research with Large Language Models," *Nature* 624, no. 7992 (2023): 570–78, https://doi.org/10.1038/s41586-023-06792-0.
[28] Kyle Swanson et al., "The Virtual Lab of AI Agents Designs New SARS-CoV-2 Nanobodies," *Nature* 646, no. 8085 (2025): 716–23, https://doi.org/10.1038/s41586-025-09442-9.
[29] Swanson et al., "The Virtual Lab of AI Agents Designs New SARS-CoV-2 Nanobodies."

Returning to the April 2025 National Academies assessment that BDTs cannot design self-replicating pathogens, a preprint study disproved this merely five months later in September 2025.[30] Scientists fine-tuned Evo 1 and Evo 2 on DNA data of bacteriophages, which is a type of virus, to determine capabilities at scale of whole genome generation.[31] Using their fine-tuned models, they designed new bacteriophage genomes and tested them on *E. coli* bacteria.[32] Many of these bacteriophages worked with some stronger than natural ones and several capable of killing resistant bacteria.[33] The statement that BDTs on their own cannot design pathogens is correct, but the key of this study was that by using BDTs as a base and fine-tuning with specific data, the BDTs transform into models that can design pathogens. However, the National Academies assessment was solely incorrect as this team leveraged existing viral sequence data to create a self-replicating pathogen. While the study is not yet peer-reviewed as a preprint, it is likely to be officially published in a journal in the coming months. Such results demonstrate that it is not a matter of whether anticipated threats will occur but when they will occur.

With the recent release of GPT-5 by OpenAI along with advancements by competitors like Google and Anthropic, the ability to automate design, lab processes, and testing has shrunk months of work to mere hours. Replicating the talents of a team of scientists shows how even a lone individual can harness such power to create a bioweapon. Paired with the discoveries in bypassing biosecurity measures, it is impossible to account for every possible vulnerability.

Strategies for Combating Bioweapons

[30] Samuel H. King et al., "Generative Design of Novel Bacteriophages with Genome Language Models," preprint, bioRxiv, September 17, 2025, 2025.09.12.675911, https://doi.org/10.1101/2025.09.12.675911.
[31] King et al., "Generative Design of Novel Bacteriophages with Genome Language Models."
[32] King et al., "Generative Design of Novel Bacteriophages with Genome Language Models."
[33] King et al., "Generative Design of Novel Bacteriophages with Genome Language Models."

DETERRENCE THEORY IN BIOSECURITY

The question then becomes what should be done to combat the dual-use of AI for bioweapons. Scholars have applied nuclear deterrence theory to biosecurity to study strategies. The key factors for effective deterrence are attribution, capability, credibility, and communication. Attribution is identifying the source of an attack. Capability is the capacity to retaliate or block an attack. Credibility is the willingness to act as perceived by an adversary. Communication is clearly conveying threats, intentions, and limitations to an adversary, which influence credibility. Based on these, three camps have formed as illustrated in Figure 2: deterrence by punishment, denial, and dissuasion.

DETERRENCE BY PUNISHMENT IN BIOSECURITY. Deterrence by punishment involves threatening adversaries with significant retaliation if they attack first. Attribution is notoriously difficult for bioweapons as demonstrated with determining the exact origin of COVID-19, which is still rigorously debated.[34] It is difficult to determine how, where, and when an infection occurs due to the nature of pathogens. To punish an attack, the inflicted party must know the identity of the adversary. Without attribution, punishment is infeasible.

There have been attempts at addressing the attribution issue. Propositions largely center on improving current investigation mechanisms or creating new ones.[35] More recently, researchers

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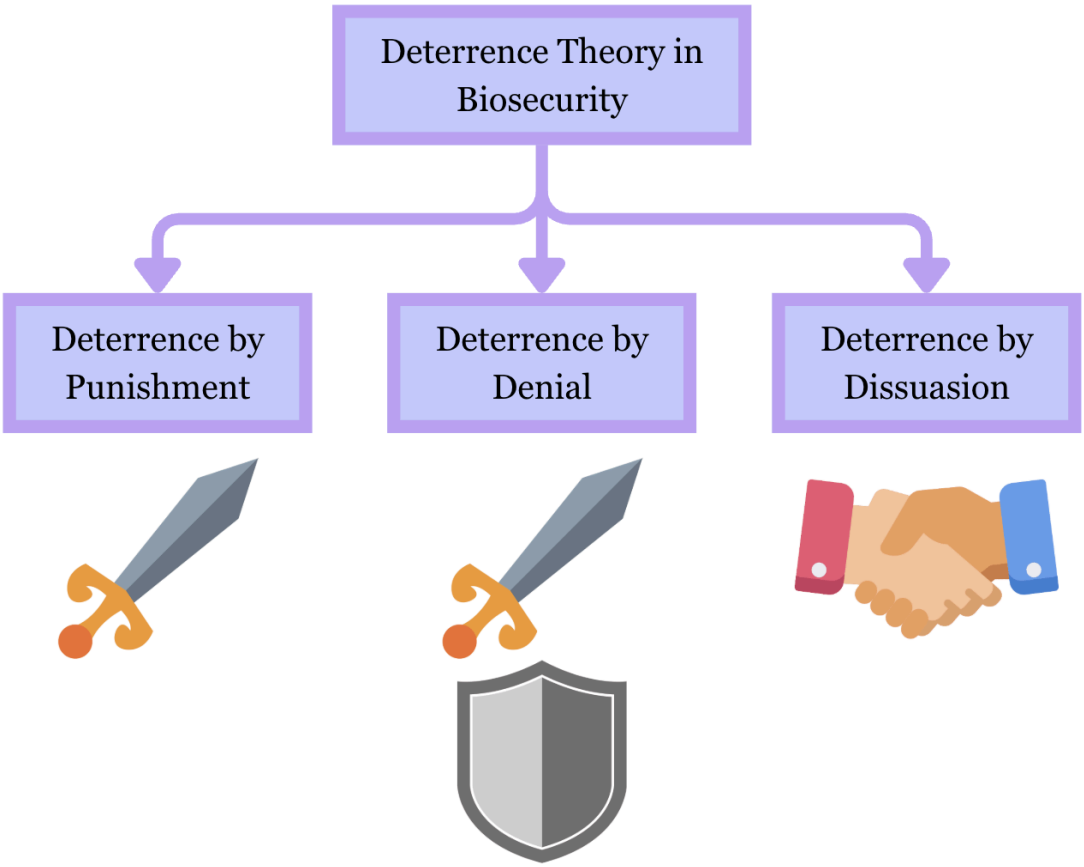
[34] Jaime M. Yassif et al., "Guarding Against Catastrophic Biological Risks: Preventing State Biological Weapon Development and Use by Shaping Intentions," *Health Security* 21, no. 4 (2023): 258–65, https://doi.org/10.1089/hs.2022.0145; Nathan Paxton et al., "Disincentivizing Bioweapons: Theory and Policy Approaches," *The Nuclear Threat Initiative*, December 10, 2024, https://www.nti.org/analysis/articles/disincentivizing-bioweapons-theory-and-policy-approaches/.
[35] Yassif et al., "Guarding Against Catastrophic Biological Risks"; Angela Kane et al., "Joint Assessment Mechanism to Determine Pandemic Origins," *The Nuclear Threat Initiative*, March 13, 2024, https://www.nti.org/about/programs-projects/project/joint-assessment-mechanism-to-determine-pandemic-origins/.

Figure 2. The Three Camps of Deterrence Theory in Biosecurity



have turned to AI. One algorithm called deteRNNt identifies signatures in genetic sequences and reached 70% attribution accuracy on over 1,300 labs.[36] However, this is still limited, especially if the adversary is off-the-grid or uses AI for generation, in which such a sequence may not be attributable to the particular adversary. The one useful application is for accidental lab leaks, which is irrelevant to deterrence.

DETERRENCE BY DENIAL IN BIOSECURITY. Deterrence by denial involves disincentivizing adversaries by making success unlikely and denying desired benefits.[37] This extends upon

[36] Ethan C. Alley et al., "A Machine Learning Toolkit for Genetic Engineering Attribution to Facilitate Biosecurity," *Nature Communications* 11, no. 1 (2020): 6293, https://doi.org/10.1038/s41467-020-19612-0.
[37] Yassif et al., "Guarding Against Catastrophic Biological Risks."

punishment by including immense defensive capabilities. In biosecurity, this makes attribution less necessary but capability much harder to achieve due to the dual cost of developing working offensive and defensive mechanisms.[38] Denial assumes that the threat already exists with potential adversaries possessing or intending to develop bioweapons and new technologies lowering the skill threshold required.

Strong deterrence by denial requires acknowledging the different motivations between state and nonstate actors.[39] State actors often have enough financial resources and trained individuals, which may require making the costs exceedingly high to deter.[40] Nonstate actors are usually not motivated by the same military, economic, or political goals as states and may simply possess a desire to cause catastrophic harm.[41]

Research on denial strategy focuses extensively on policy.[42] For example, strengthening private-public partnerships can increase biomanufacturing capabilities, allowing for faster vaccine or treatment production.[43] Not only does this deter bioweapons development, but this also allows more rapid response to natural pathogens. Like for punishment, recommendations also include improving investigative bodies and instituting accountability systems to support the BWC.[44]

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[38] Paxton et al., "Disincentivizing Bioweapons."
[39] Yassif et al., "Guarding Against Catastrophic Biological Risks."
[40] Yassif et al., "Guarding Against Catastrophic Biological Risks."
[41] Yassif et al., "Guarding Against Catastrophic Biological Risks."
[42] Yassif et al., "Guarding Against Catastrophic Biological Risks"; Paxton et al., "Disincentivizing Bioweapons"; Bloomfield et al., "AI and Biosecurity"; Wang et al., "A Call for Built-in Biosecurity Safeguards for Generative AI Tools."
[43] Shannon Green et al., "A Biomanufacturing Plan to Confront Future Biological Threats - The Council on Strategic Risks," September 4, 2024, https://councilonstrategicrisks.org/2024/09/04/a-biomanufacturing-plan-to-confront-future-biological-threats/.
[44] Yassif et al., "Guarding Against Catastrophic Biological Risks."

DETERRENCE BY DISSUASION IN BIOSECURITY. Deterrence by dissuasion involves convincing an adversary not to develop bioweapons by showing the costs greatly outweigh the expected benefit.[45] This strategy centers on eliminating the threat before it exists. The costs of capability are avoided by mainly relying on diplomatic and political means.[46]

In the past, dissuasion has been the more realistic option. An issue shared by denial and punishment is what communication entails. By communicating capability to retaliate or defend, a state implicitly justifies developing bioweapons or suggests it already has a bioweapons program, both risking proliferation. Historical cases show that even if state or nonstate actors have intent, they lack the means, particularly due to limited expertise, which undermines the argument for denial.[47] Since state actors then become the only possible capable entity, supporters of dissuasion argue that the only communication required is that bioweapons can easily spread and infect their own leaders and citizens, which disincentives any country with some concern for its own population.[48] Evidence of generative AI models only providing slight advantages compared to using the Internet also bolstered support for dissuasion.[49]

DETERRENCE THEORY IN BIOSECURITY TODAY

[45] Paxton et al., "Disincentivizing Bioweapons"; National Academies of Sciences, Engineering, and Medicine, *Assessing and Improving Strategies for Preventing, Countering, and Responding to Weapons of Mass Destruction Terrorism: Biological Threats: Proceedings of a Workshop–in Brief* (The National Academies Press, 2023), https://doi.org/10.17226/26845.
[46] Paxton et al., "Disincentivizing Bioweapons."
[47] Paxton et al., "Disincentivizing Bioweapons"; Sonia Ben Ouagrham-Gormley, "Barriers to Bioweapons: Intangible Obstacles to Proliferation," *International Security* 36, no. 4 (2012): 80–114, https://doi.org/10.1162/ISEC_a_00077.
[48] Paxton et al., "Disincentivizing Bioweapons."
[49] Paxton et al., "Disincentivizing Bioweapons."

AI today fundamentally shifts the dynamics between the three camps. Punishment remains the weakest due to the challenge of attribution, especially when a decisive and quick response is needed for effective deterrence. By the time a bioweapon has been identified and attributed, damage would have likely already been done. While dissuasion may have been the best option, the most recent findings of AI's capabilities as discussed previously completely dismantle the argument for it. There is practically no expertise required, and AI can develop pathogens. This means even if a state actor can be reasoned with through communication, a nonstate actor, which may have the goal of global catastrophe, cannot. With the barriers to entry removed, there is essentially nothing stopping an actor determined to unleash bioweapons. By process of elimination, deterrence by denial is the only option left, necessitating an impenetrable biodefense infrastructure.

AI for Defensive Biosecurity

Policy changes alone will not do much to mitigate the bioweapons threat. AI advances too rapidly, leaving existing frameworks, institutions, and policies unable to adapt quickly enough. One possible solution is to leverage AI innovations to our advantage through biodefense.

There is little research done on AI as a defense mechanism for biosecurity. One application is for genomic surveillance to monitor outbreaks by leveraging protein language models, which can predict viral evolution and properties.[50] Another application is utilizing demonstrated automation

[50] Spyros Lytras et al., "Pathogen Genomic Surveillance and the AI Revolution," *Journal of Virology* 99, no. 2 (2025): e01601-24, https://doi.org/10.1128/jvi.01601-24.

and acceleration of lab processes and experiments in industries like biomanufacturing to increase their current capacity and ability to act with speed.

All of the research on AI applications for biodefense focus on an institutional or system level application, which still allows for significant response latency. This includes the time from a patient's infection to diagnosis and from pathogen identification to the initial research required to understand biological underpinnings before designing pharmaceutical treatments. Ultimately, the frontline of biosecurity occurs at the patient level due to the nature of how pathogens spread. There is currently no research done on how AI can function on a direct patient level as frontline biodefense.

Conclusion

This literature review emphasized six major takeaways regarding the intersection of AI and biosecurity. First, AI enables all actors to create bioweapons with little expertise at affordable costs. Second, AI will only become more powerful with new unknown vulnerabilities. Third, deterrence by denial remains the only feasible strategy for bioweapons. Fourth, policy changes alone cannot keep pace with AI advancements. Fifth, there is scarce research on AI for biodefense with frontline, patient level applications particularly neglected. Bridging the gap between international security theory and AI is essential to advance into a new phase of biosecurity.

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